# Supplementary Material for "CascadeRAG: Abstention-Triggered Escalation that Outperforms Costlier Retrieval-Augmented Generation Architectures at a Fraction of their Cost"

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## S1 Full per-set results

Recall is measured over the passages actually in the generator's context, whose mean size is reported beside it (Section 6.3 of the article); "EM answ." is Exact Match restricted to the questions the system did not decline. Dashes mark metrics that are undefined for a row (ClosedBook retrieves nothing; NaturalQuestions and TriviaQA annotate no gold documents, so answer-in-context replaces Recall there).

### S1.1 MuSiQue

| Architecture | EM | F1 | Abst. | EM answ. | Recall | Passages | Calls |
|---|---|---|---|---|---|---|---|
| ClosedBook | 0.0381 | 0.1186 | 0.1320 | 0.0439 | – | – | 1.00 |
| BM25 | 0.0546 | 0.1162 | 0.3037 | 0.0784 | 0.2114 | 5.00 | 1.00 |
| NaiveRAG-5 | 0.0596 | 0.1322 | 0.2607 | 0.0806 | 0.2209 | 5.00 | 1.00 |
| NaiveRAG-10 | 0.0728 | 0.1532 | 0.2098 | 0.0921 | 0.2666 | 10.00 | 1.00 |
| NaiveRAG-12 | 0.0712 | 0.1503 | 0.1816 | 0.0870 | 0.2812 | 12.00 | 1.00 |
| Hybrid | 0.0695 | 0.1462 | 0.2459 | 0.0922 | 0.2616 | 5.00 | 1.00 |
| HyDE | 0.0753 | 0.1596 | 0.2334 | 0.0983 | 0.2420 | 5.00 | 2.00 |
| Rerank | 0.0811 | 0.1634 | 0.2127 | 0.1030 | 0.2678 | 5.00 | 1.00 |
| Neighbour | 0.0691 | 0.1379 | 0.2205 | 0.0886 | 0.2209 | 12.26 | 1.00 |
| CRAG-5 | 0.0832 | 0.1634 | 0.2315 | 0.1083 | 0.2326 | 5.35 | 2.99 |
| CRAG-10 | 0.0877 | 0.1711 | 0.1953 | 0.1090 | 0.2732 | 10.00 | 3.00 |
| IRCoT | 0.0695 | 0.1478 | 0.2441 | 0.0920 | 0.2477 | 7.35 | 2.54 |
| Adaptive | 0.0666 | 0.1410 | 0.2482 | 0.0886 | 0.2365 | 6.64 | 2.96 |
| Cascade | 0.0840 | 0.1713 | 0.0335 | 0.0869 | 0.2391 | 4.41 | 1.58 |

## S1.2 HotpotQA

| Architecture | EM | F1 | Abst. | EM answ. | Recall | Passages | Calls |
|---|---|---|---|---|---|---|---|
| ClosedBook | 0.1887 | 0.2740 | 0.0204 | 0.1926 | – | – | 1.00 |
| BM25 | 0.3063 | 0.4096 | 0.1164 | 0.3466 | 0.4810 | 5.00 | 1.00 |
| NaiveRAG-5 | 0.3248 | 0.4313 | 0.1124 | 0.3659 | 0.5405 | 5.00 | 1.00 |
| NaiveRAG-10 | 0.3299 | 0.4367 | 0.0905 | 0.3627 | 0.5994 | 10.00 | 1.00 |
| NaiveRAG-12 | 0.3363 | 0.4448 | 0.0743 | 0.3632 | 0.6126 | 12.00 | 1.00 |
| Hybrid | 0.3368 | 0.4462 | 0.0912 | 0.3706 | 0.5646 | 5.00 | 1.00 |
| HyDE | 0.3338 | 0.4434 | 0.0974 | 0.3698 | 0.5552 | 5.00 | 2.00 |
| Rerank | 0.3757 | 0.4911 | 0.0791 | 0.4080 | 0.6408 | 5.00 | 1.00 |
| Neighbour | 0.3190 | 0.4251 | 0.0940 | 0.3521 | 0.5405 | 11.34 | 1.00 |
| CRAG-5 | 0.3284 | 0.4367 | 0.1035 | 0.3663 | 0.5462 | 5.32 | 2.99 |
| CRAG-10 | 0.3410 | 0.4505 | 0.0821 | 0.3715 | 0.6017 | 10.00 | 3.00 |
| IRCoT | 0.3409 | 0.4493 | 0.1030 | 0.3800 | 0.5629 | 6.13 | 2.31 |
| Adaptive | 0.3302 | 0.4378 | 0.1093 | 0.3707 | 0.5471 | 5.29 | 2.30 |
| Cascade | 0.3583 | 0.4713 | 0.0043 | 0.3598 | 0.5793 | 4.79 | 1.22 |

## S1.3 2WikiMultihopQA

| Architecture | EM | F1 | Abst. | EM answ. | Recall | Passages | Calls |
|---|---|---|---|---|---|---|---|
| ClosedBook | 0.2455 | 0.2969 | 0.0223 | 0.2511 | – | – | 1.00 |
| BM25 | 0.2614 | 0.3222 | 0.1561 | 0.3097 | 0.3730 | 5.00 | 1.00 |
| NaiveRAG-5 | 0.2285 | 0.2974 | 0.1672 | 0.2744 | 0.3540 | 5.00 | 1.00 |
| NaiveRAG-10 | 0.2522 | 0.3196 | 0.1269 | 0.2888 | 0.3881 | 10.00 | 1.00 |
| NaiveRAG-12 | 0.2516 | 0.3198 | 0.1172 | 0.2850 | 0.3960 | 12.00 | 1.00 |
| Hybrid | 0.2696 | 0.3338 | 0.1374 | 0.3125 | 0.4081 | 5.00 | 1.00 |
| HyDE | 0.2449 | 0.3132 | 0.1573 | 0.2906 | 0.3686 | 5.00 | 2.00 |
| Rerank | 0.2675 | 0.3311 | 0.1273 | 0.3065 | 0.3941 | 5.00 | 1.00 |
| Neighbour | 0.2395 | 0.3083 | 0.1483 | 0.2812 | 0.3542 | 10.72 | 1.00 |
| CRAG-5 | 0.2332 | 0.3014 | 0.1808 | 0.2846 | 0.3375 | 5.17 | 3.00 |
| CRAG-10 | 0.2535 | 0.3205 | 0.1350 | 0.2930 | 0.3758 | 10.00 | 3.00 |
| IRCoT | 0.2467 | 0.3165 | 0.1556 | 0.2921 | 0.3887 | 6.96 | 2.48 |
| Adaptive | 0.2289 | 0.2978 | 0.1678 | 0.2750 | 0.3560 | 5.14 | 2.11 |
| Cascade | 0.2783 | 0.3436 | 0.0052 | 0.2797 | 0.3808 | 4.67 | 1.32 |

## S1.4 NaturalQuestions

| Architecture | EM | F1 | Abst. | EM answ. | Ans. in ctx | Passages | Calls |
|---|---|---|---|---|---|---|---|
| ClosedBook | 0.1399 | 0.2374 | 0.0823 | 0.1524 | – | – | 1.00 |
| BM25 | 0.2249 | 0.3166 | 0.2349 | 0.2940 | 0.4559 | 5.00 | 1.00 |
| NaiveRAG-5 | 0.3349 | 0.4436 | 0.1693 | 0.4031 | 0.7262 | 5.00 | 1.00 |
| Hybrid | 0.3089 | 0.4228 | 0.1748 | 0.3743 | 0.6707 | 5.00 | 1.00 |
| HyDE | 0.3476 | 0.4602 | 0.1598 | 0.4138 | 0.7525 | 5.00 | 2.00 |
| Rerank | 0.2352 | 0.3177 | 0.2773 | 0.3254 | 0.5097 | 5.00 | 1.00 |
| CRAG-5 | 0.3415 | 0.4529 | 0.1660 | 0.4094 | 0.7554 | 5.53 | 2.94 |
| IRCoT | 0.3357 | 0.4451 | 0.1657 | 0.4024 | 0.7298 | 5.29 | 2.09 |
| Adaptive | 0.3352 | 0.4433 | 0.1704 | 0.4040 | 0.7253 | 5.01 | 2.05 |
| Cascade | 0.3417 | 0.4603 | 0.0316 | 0.3528 | 0.6748 | 4.52 | 1.44 |

### S1.5 TriviaQA

| Architecture | EM | F1 | Abst. | EM answ. | Ans. in ctx | Passages | Calls |
|---|---|---|---|---|---|---|---|
| ClosedBook | 0.4241 | 0.4975 | 0.0147 | 0.4304 | – | – | 1.00 |
| BM25 | 0.5356 | 0.6220 | 0.0994 | 0.5947 | 0.6706 | 5.00 | 1.00 |
| NaiveRAG-5 | 0.5700 | 0.6594 | 0.0796 | 0.6193 | 0.7358 | 5.00 | 1.00 |
| Hybrid | 0.5835 | 0.6732 | 0.0666 | 0.6252 | 0.7494 | 5.00 | 1.00 |
| HyDE | 0.5954 | 0.6854 | 0.0665 | 0.6378 | 0.7681 | 5.00 | 2.00 |
| Rerank | 0.5962 | 0.6847 | 0.0631 | 0.6364 | 0.7625 | 5.00 | 1.00 |
| CRAG-5 | 0.5845 | 0.6739 | 0.0720 | 0.6298 | 0.7560 | 5.39 | 2.97 |
| IRCoT | 0.5723 | 0.6623 | 0.0756 | 0.6190 | 0.7416 | 5.38 | 2.11 |
| Adaptive | 0.5698 | 0.6593 | 0.0791 | 0.6187 | 0.7346 | 5.02 | 2.06 |
| Cascade | 0.6032 | 0.6933 | 0.0044 | 0.6059 | 0.7617 | 4.84 | 1.16 |

### S1.6 Cascade ablations (Exact Match)

| Variant | MuSiQue | HotpotQA | 2Wiki | NQ | TriviaQA |
|---|---|---|---|---|---|
| Stage 1 alone (fused, no escalation) | 0.0783 | 0.3521 | 0.2630 | 0.3257 | 0.5921 |
| Cascade (proposed) | 0.0840 | 0.3583 | 0.2783 | 0.3417 | 0.6032 |
| Cascade, rerank first | 0.0840 | 0.3825 | 0.2823 | 0.2951 | 0.6079 |
| Cascade, naive first | 0.0662 | 0.3353 | 0.2542 | 0.3474 | 0.5869 |
| Cascade, hybrid first | 0.0757 | 0.3456 | 0.2819 | 0.3269 | 0.5973 |
| Cascade + confidence trigger | 0.0841 | 0.3562 | 0.2745 | 0.3408 | 0.6053 |
| IRCoT, one-shot | 0.0712 | 0.3367 | 0.2565 | 0.3358 | 0.5756 |

### S1.7 Second reader (Llama-3.1-8B-Instruct, Exact Match)

| Architecture | MuSiQue | HotpotQA | 2Wiki | NQ | TriviaQA |
|---|---|---|---|---|---|
| ClosedBook | 0.0451 | 0.2182 | 0.2932 | 0.2463 | 0.5839 |
| NaiveRAG-5 | 0.0728 | 0.3355 | 0.2228 | 0.3493 | 0.5885 |
| Rerank | 0.1031 | 0.3870 | 0.2603 | 0.2617 | 0.6100 |
| Stage 1 alone (fused) | 0.0869 | 0.3633 | 0.2550 | 0.3572 | 0.6086 |
| Cascade | 0.0961 | 0.3704 | 0.2770 | 0.3658 | 0.6168 |
| Cascade, calls/question | 1.41 | 1.14 | 1.29 | 1.16 | 1.08 |

| Baseline (calls/q) | NQ | TriviaQA | HotpotQA | 2Wiki | MuSiQue |
|---|---|---|---|---|---|
| ClosedBook (no retrieval) (1.0) | **+0.119** | **+0.033** | **+0.152** | -0.016 | **+0.051** |
| NaiveRAG, 5 passages (1.0) | **+0.016** | **+0.028** | **+0.035** | **+0.054** | **+0.023** |
| Rerank (cross-encoder) (1.0) | **+0.104** | +0.007 | -0.016 | **+0.017** | -0.007 |
| Fused retrieval (stage 1 alone) (1.0) | **+0.009** | **+0.008** | **+0.007** | **+0.022** | **+0.009** |
| Cascade calls/q | 1.16 | 1.08 | 1.14 | 1.29 | 1.41 |

## S2 The confidence-threshold sweep

Swept offline over 4988 sampled train questions by composing the three independently-run stages question by question (Section 5.3 of the article). “Escalated” counts stage-1 answers handed on at that threshold, “discarded” the correct answers among them, “missed” those no later stage recovered. Exact Match peaks at 0.3941 (threshold -0.150) against 0.3923 for the abstention-only cascade.

| Threshold | EM | F1 | Calls | Escalated | Discarded | Missed |
|---|---|---|---|---|---|---|
| $-\infty$ (abstention only) | 0.3923 | 0.4708 | 1.35 | 687 | 0 | 0 |
| -2.000 | 0.3923 | 0.4708 | 1.35 | 687 | 0 | 0 |
| -1.500 | 0.3923 | 0.4708 | 1.35 | 688 | 0 | 0 |
| -1.000 | 0.3923 | 0.4708 | 1.35 | 697 | 2 | 1 |
| -0.800 | 0.3923 | 0.4707 | 1.37 | 723 | 7 | 2 |
| -0.600 | 0.3921 | 0.4701 | 1.38 | 755 | 16 | 3 |
| -0.500 | 0.3925 | 0.4706 | 1.40 | 785 | 26 | 6 |
| -0.400 | 0.3937 | 0.4714 | 1.42 | 831 | 33 | 8 |
| -0.300 | 0.3939 | 0.4717 | 1.46 | 909 | 50 | 12 |
| -0.200 | 0.3917 | 0.4701 | 1.53 | 1046 | 93 | 33 |
| -0.150 | 0.3941 | 0.4714 | 1.60 | 1169 | 115 | 41 |
| -0.100 | 0.3941 | 0.4722 | 1.72 | 1385 | 159 | 62 |
| -0.050 | 0.3939 | 0.4701 | 1.94 | 1764 | 216 | 93 |
| -0.020 | 0.3919 | 0.4662 | 2.20 | 2194 | 312 | 139 |
| -0.010 | 0.3917 | 0.4653 | 2.36 | 2468 | 378 | 167 |
| -0.005 | 0.3865 | 0.4601 | 2.50 | 2691 | 464 | 208 |

# S3 Prompts

All prompts, verbatim from the implementation. The reader prompt is shared by every architecture; the instruction to output only the answer exists because the gold answers are short spans (Section 6.3 of the article).

**Reader (all architectures).**

```
Answer the question based on the given documents.  Only give me the answer
and do not output any other words.
Documents:
[1] ... [k] ...
Question:  {question}
```

**ClosedBook reader.** As above, minus the documents block.

**HyDE drafter.**

```
Write a short encyclopedia passage that answers the question.
Question:  {question}
Name the people, places, dates and titles involved.  Write it as an
encyclopedia entry, not as a reply to the question.  If you are not sure of
the facts, write the most plausible passage anyway.  Reply with only the
passage.
```

**IRCoT reasoning step.**

```
You are answering a question step by step using the documents below.
Documents:  ...
Question:  {question}
Reasoning so far:  ...
Write the next single sentence of reasoning.  Name the entities that still
need to be looked up.  If the documents already answer the question, write
"So the answer is:  <answer>".
```

The one-shot variant prepends a single hand-written two-hop worked example (identical for every question set).

**CRAG retrieval grader.**

```
Judge which documents contain information useful for answering the question.
Documents:  ...
Question:  {question}
Reply with only the numbers of the useful documents, separated by commas.
If none of them are useful, reply with "none".
```

**CRAG query rewrite.**

```
The search results for this question were not useful.
Question:  {question}
Write a better search query for finding the documents that answer it.  Spell
out the entities the question only refers to indirectly.  Reply with only
the query.
```

**Adaptive-RAG router.**

```
Classify how much looking up the question needs.
Question:  {question}
A - a widely known fact that needs no source, such as a famous person,
country or date.
B - one fact that has to be looked up in a single document.
C - several facts from different documents that have to be combined, for
example when the question refers to an entity it does not name.
Reply with only the letter A, B or C.
```

**Abstention phrases.** An answer is counted as an abstention if, after normalisation, it contains any of: *do(es) not contain, no information, not mentioned, can(not) answer, not provided, no mention, unable to answer, not enough information, insufficient information, there is no, not possible to determine* — or if it is longer than twenty words.

# References